BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 252 · Fetch–decode–execute and buses

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which register holds the address accessed during a fetch?

Q2. What does the CIR hold?

Q3. What kind of information travels along the address bus?

Q4. A rising-edge D flip-flop has Q=0. D changes to 1 between clock edges; the next rising edge occurs while D=1. State Q before and after that edge, and explain why.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

fetch — Retrieve next instruction from memory.

decode — Interpret opcode and operands.

execute — Perform the operation.

address bus — Transfers memory/device addresses.

data bus — Transfers data and instructions.

control bus — Transfers signals coordinating reads and writes.

New learning

Fetch uses the buses — The address bus carries the address selected by MAR; memory returns instruction data on the data bus; control signals coordinate the transfer.

Decode interprets instruction — The control unit examines CIR to decide which operation and operands are involved.

Execute changes state — An instruction may update ACC or memory, or alter PC for a jump. Interrupt handling is coordinated at appropriate points and can temporarily redirect execution.

Worked example

Fetch the instruction at address 40 and explain what each bus contributes.

1. Predict first — MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

2. Explain the answer — The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

We Do · Practise with guidance

An instruction at address 81 is an unconditional jump to 120. What happens to PC?

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Fetch at 81: PC advances to 82.
Execute JMP 120: PC becomes 120.
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Predict first

The initial fetch advances PC from 81 to 82, as in a sequential instruction.

Explain the answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Executing the unconditional branch overwrites PC with target address 120. The next fetch occurs there, not at 82.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For each bus—address, data and control—state its main function and give one example of the information transported.

Question 2

Trace a fetch starting PC=64, then explain what happens when CIR contains JMP 100.

Question 3

Compare what happens in the instruction cycle during a normal arithmetic instruction and while responding to an interrupt.

Exit Ticket · Answer before checking

An unconditional branch can change the Program Counter away from its sequential value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. MAR.

Q2. The current instruction that is being decoded or executed.

Q3. Addresses identifying the memory or device location.

Q4. Before edge Q stays 0; at rising edge D=1 is sampled so Q becomes 1. It changes on the active edge, not merely when D changes.

You Do pages · Independent questions

Q1. [6] Address bus identifies memory/device location [2]; data bus transports instruction/data contents [2]; control bus carries read/write/timing signals [2].

Q2. [6] PC64→MAR64 [1], instruction memory→MDR [1], MDR→CIR [1], sequential PC→65 [1], decode JMP100 [1], execute branch sets PC=100 [1].

Q3. [6] Both use fetch/decode/execute [1]; arithmetic instruction may update ACC [1]; interrupt signals an event [1]; processor preserves enough execution state [1], transfers control to an ISR [1] and restores/resumes after service [1].

Exit Ticket · Model answer

True. The jump target becomes the next fetch address.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding